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Subject: ASD & DevOps

EXPERIMENT:8

Aim: CI/CD Pipeline to host a website on AWS.

LO4, LO6 mapped:

LO4:: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

LO6: Develop and demonstrate an end-to-end DevOps pipeline mini-project integrating all stages of DevOps from code commit to deployment and monitoring.

Theory:

AWS CI/CD Pipeline is an automated software delivery process that continuously integrates code changes, builds the application, and deploys the updated application to a target environment. CI/CD stands for Continuous Integration and Continuous Delivery/Deployment. It reduces manual effort and helps deliver software changes quickly and consistently.

In this experiment, GitHub is used as the source-code repository. Whenever changes are made to the repository, AWS CodePipeline can automatically detect the changes and start the pipeline. CodePipeline coordinates the different stages of the software delivery process.

AWS CodeBuild is used as the build stage. It retrieves the source code and executes the commands specified in the buildspec.yml file. The buildspec file defines the commands and artifacts required during the build process.

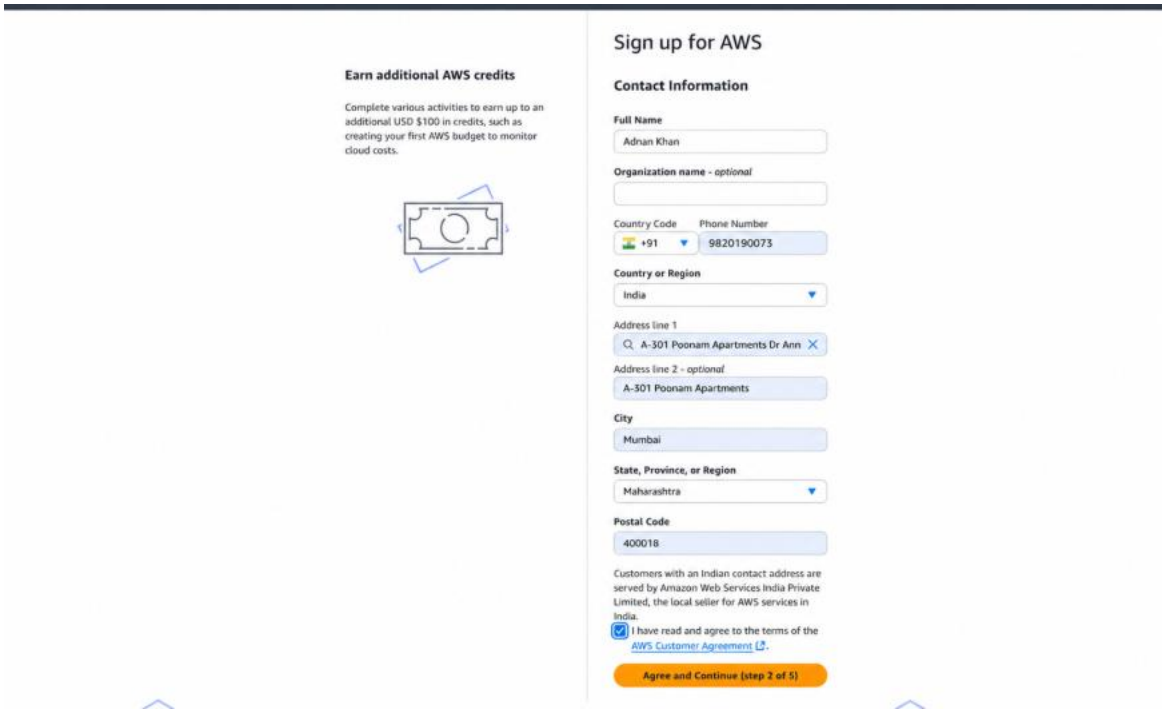
Amazon S3 is used to store the build artifacts and deploy the website. The deployed files can be hosted as a static website using S3 Static Website Hosting. An index.html file is specified as the index document, allowing the website to be accessed through the S3 website endpoint. The complete workflow implemented in this experiment is:

GitHub → AWS CodePipeline → AWS CodeBuild → Amazon S3 → Static Website

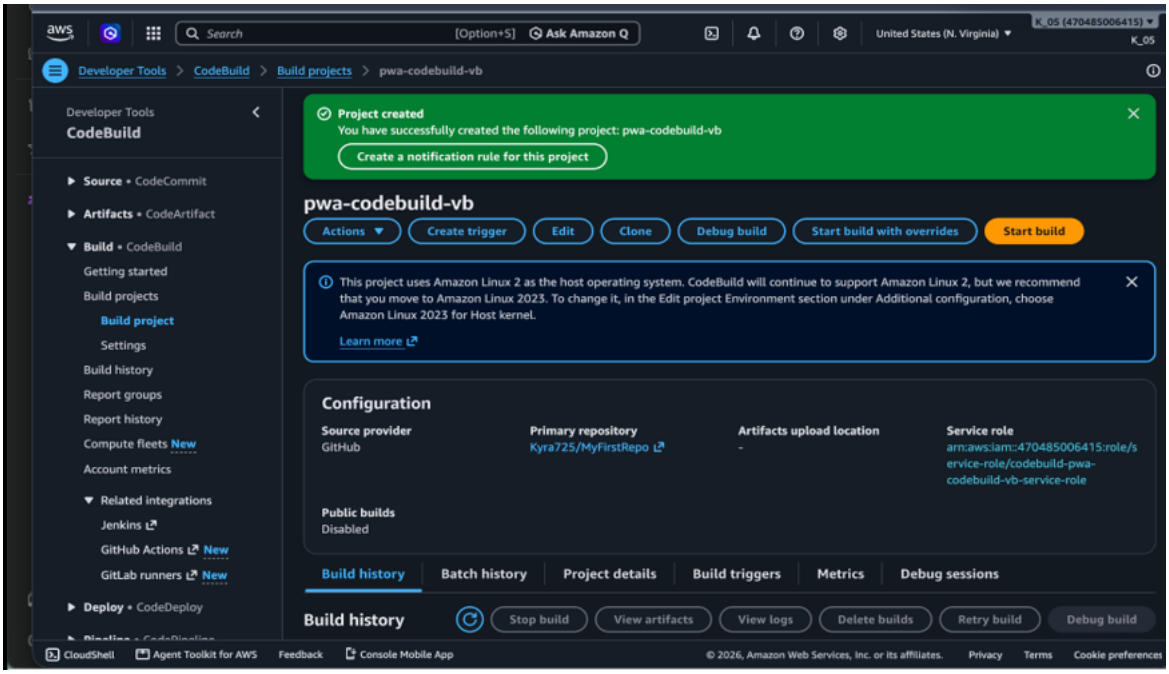
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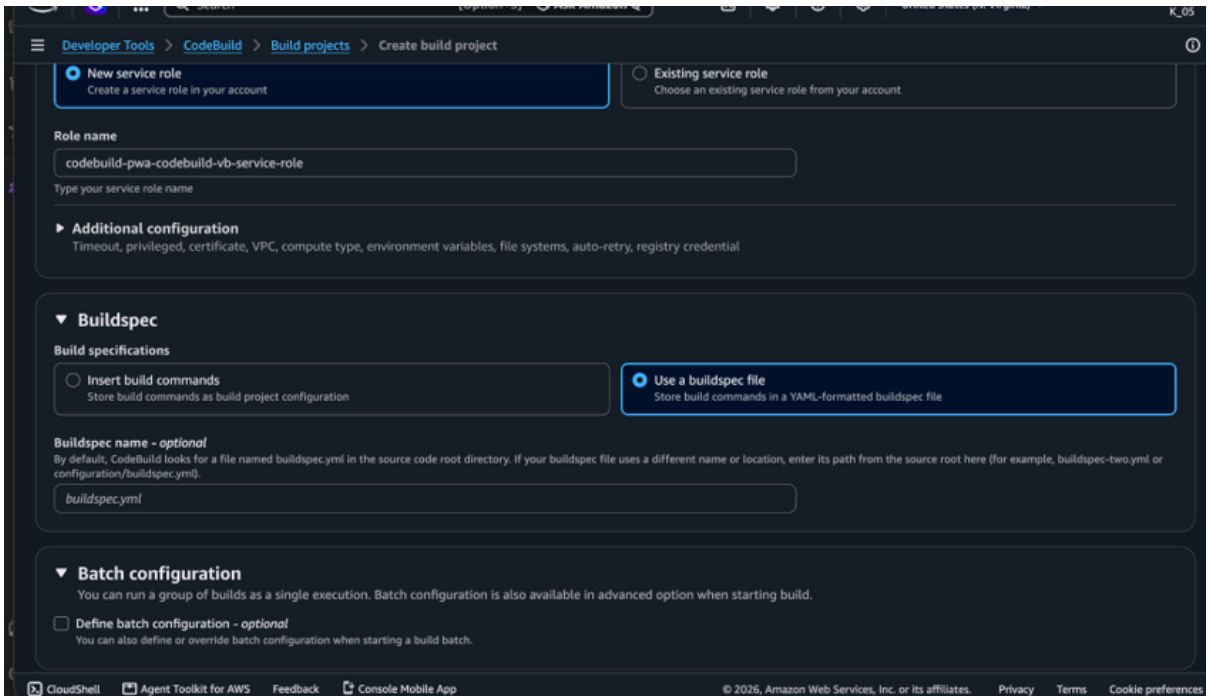
When a change is made to the source code, CodePipeline automatically triggers the required stages and updates the deployed website. This demonstrates automated deployment and continuous delivery.

Output:

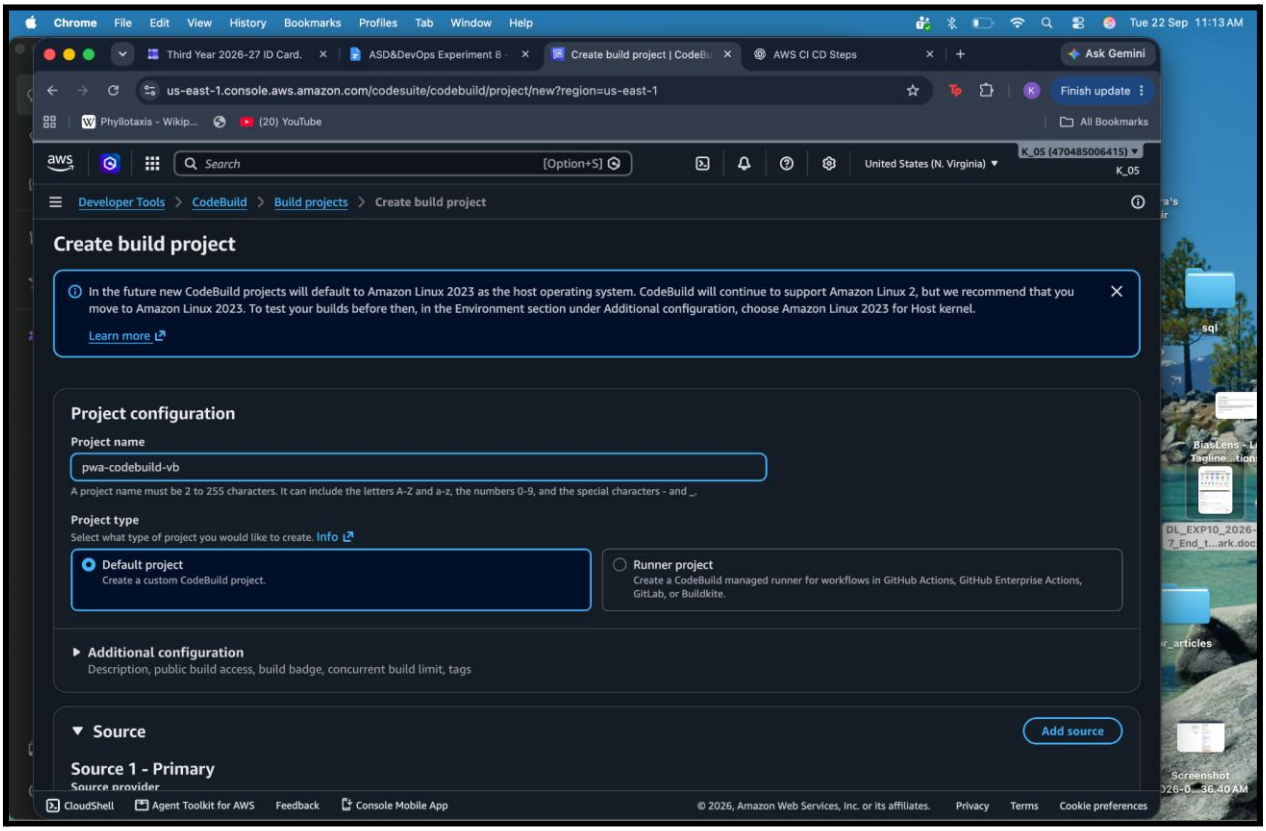


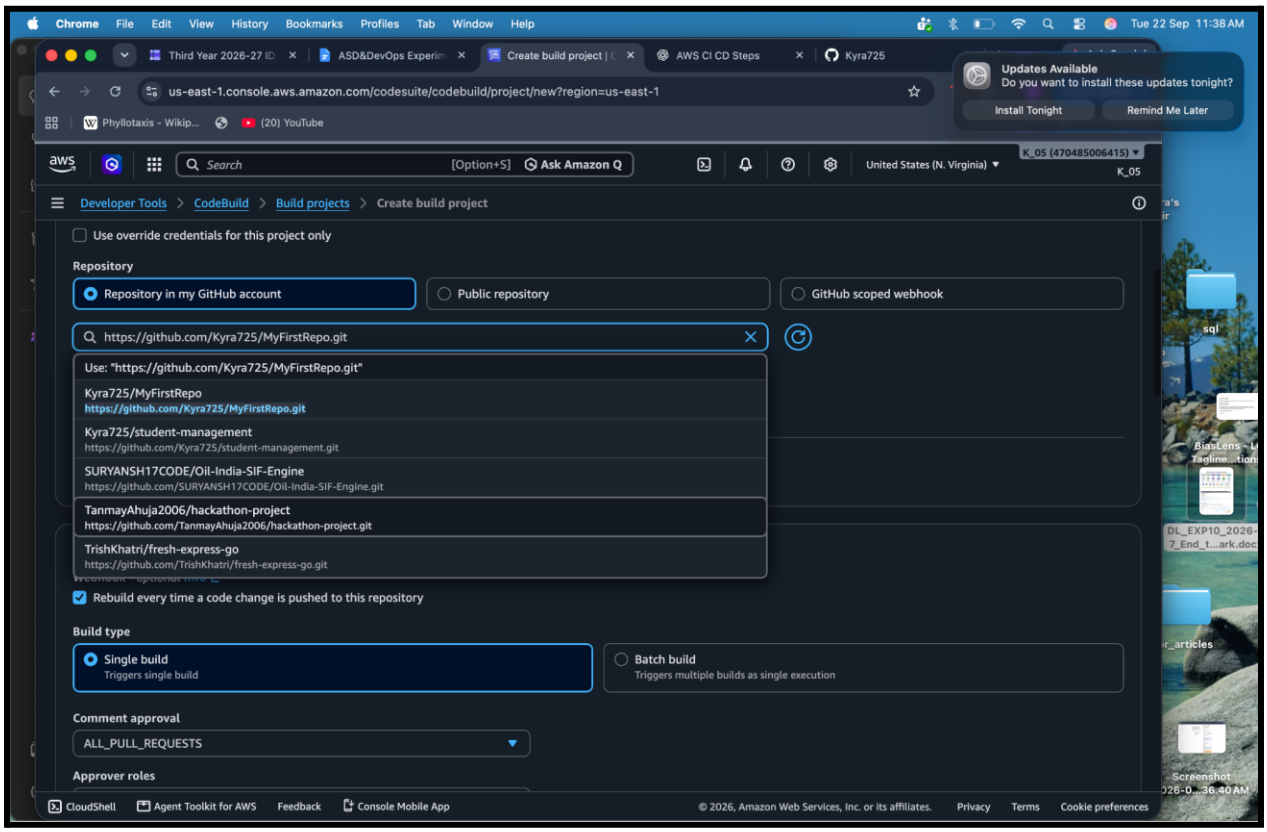
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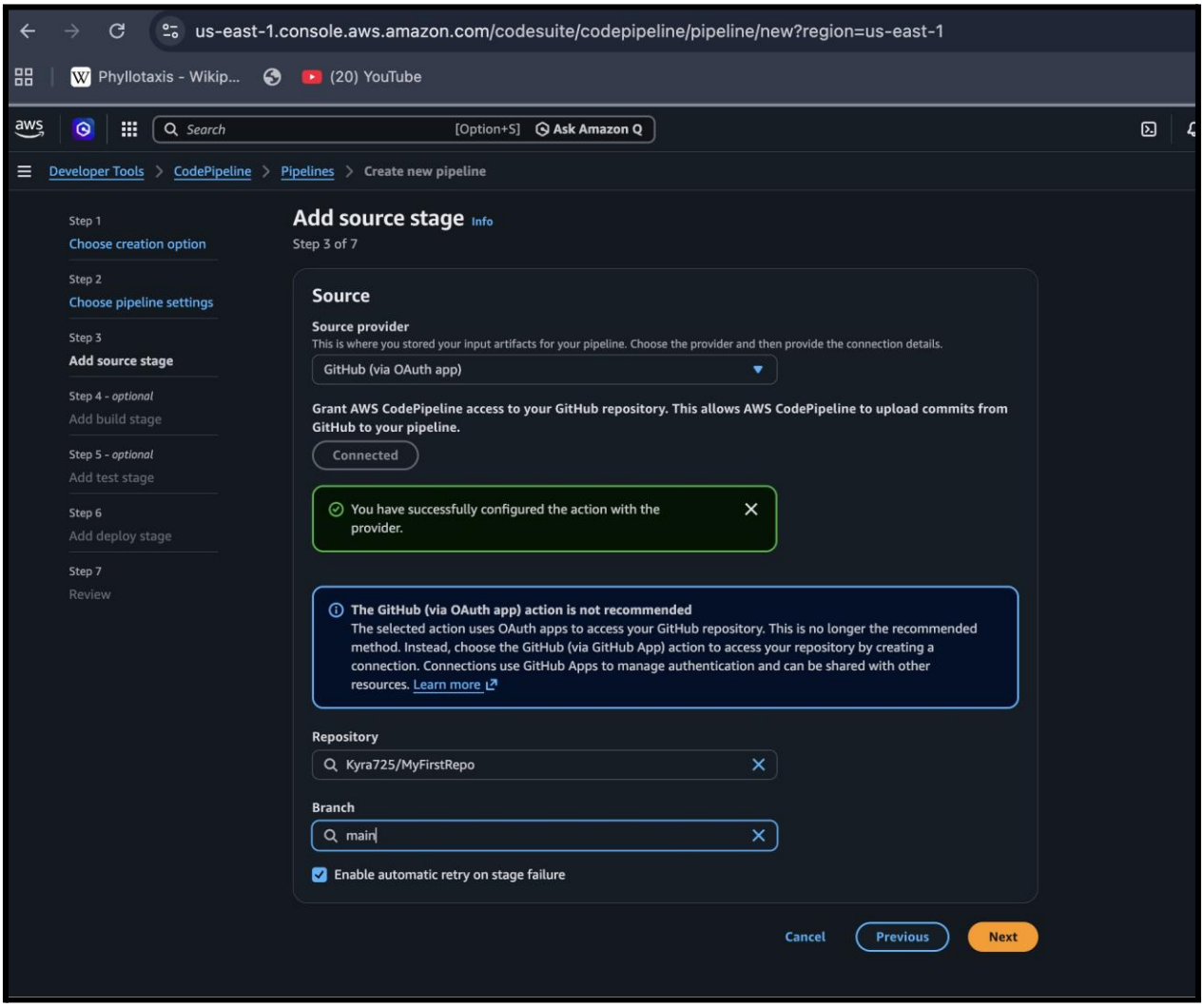


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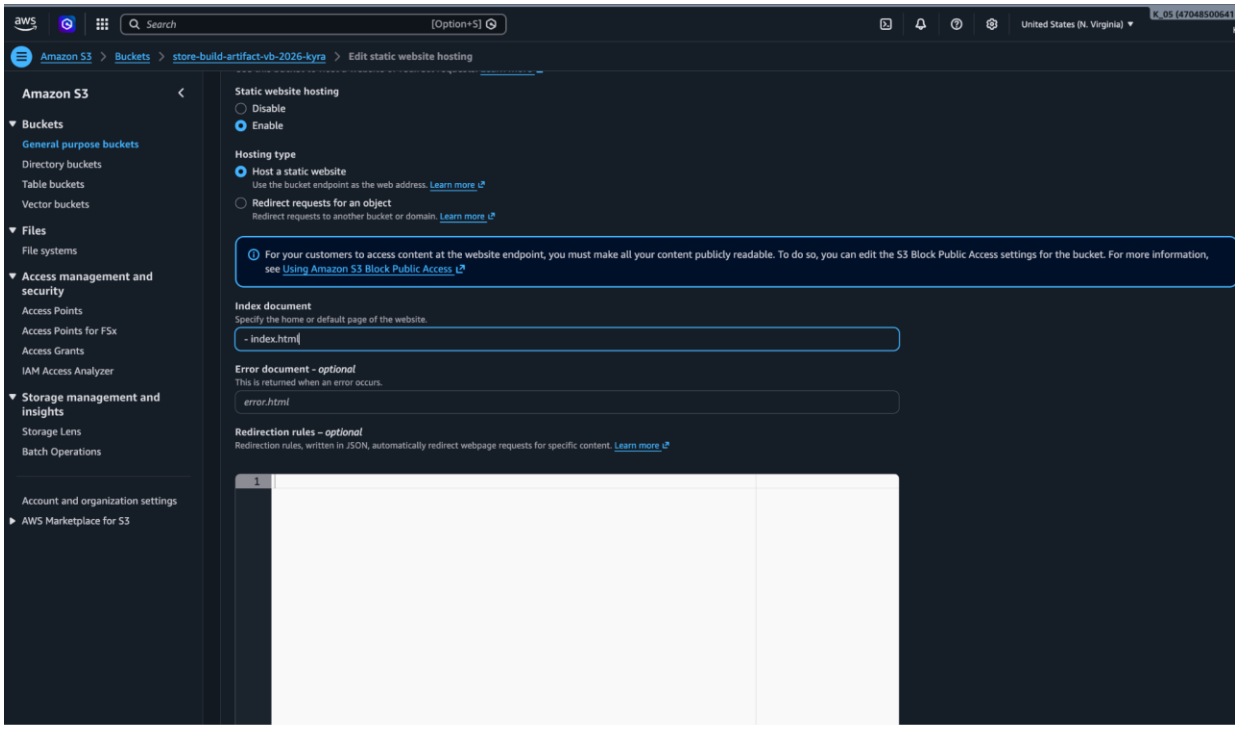
Welcome to My AWS CI/CD Website

This website is deployed automatically using AWS CodePipeline.

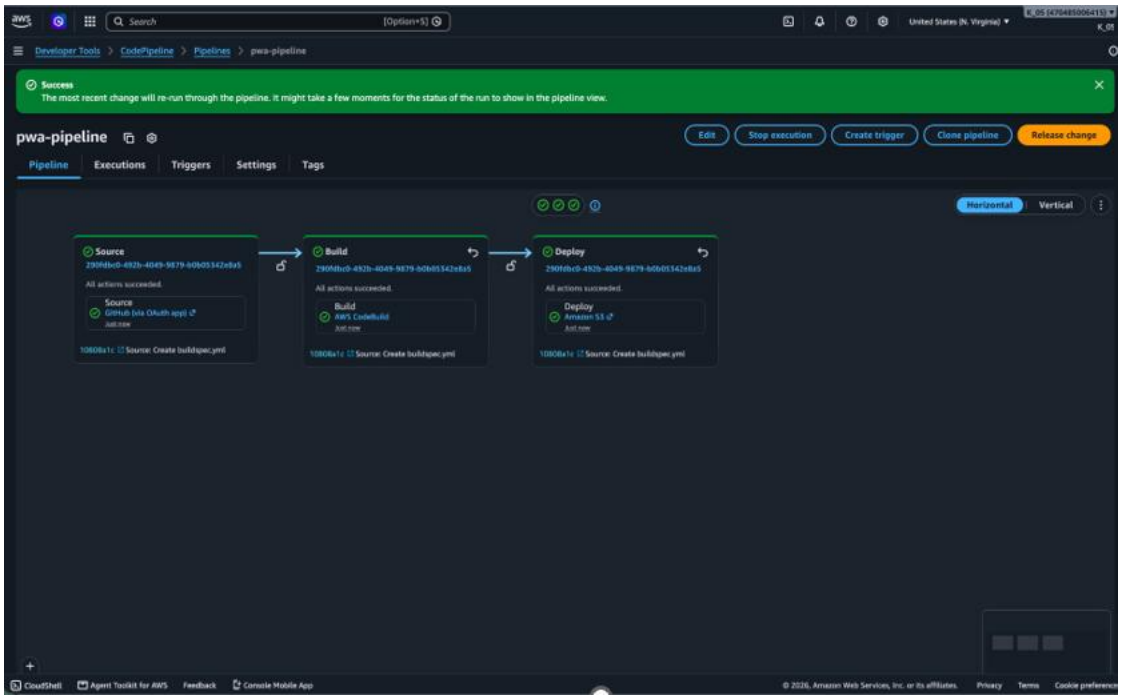
Source: GitHub

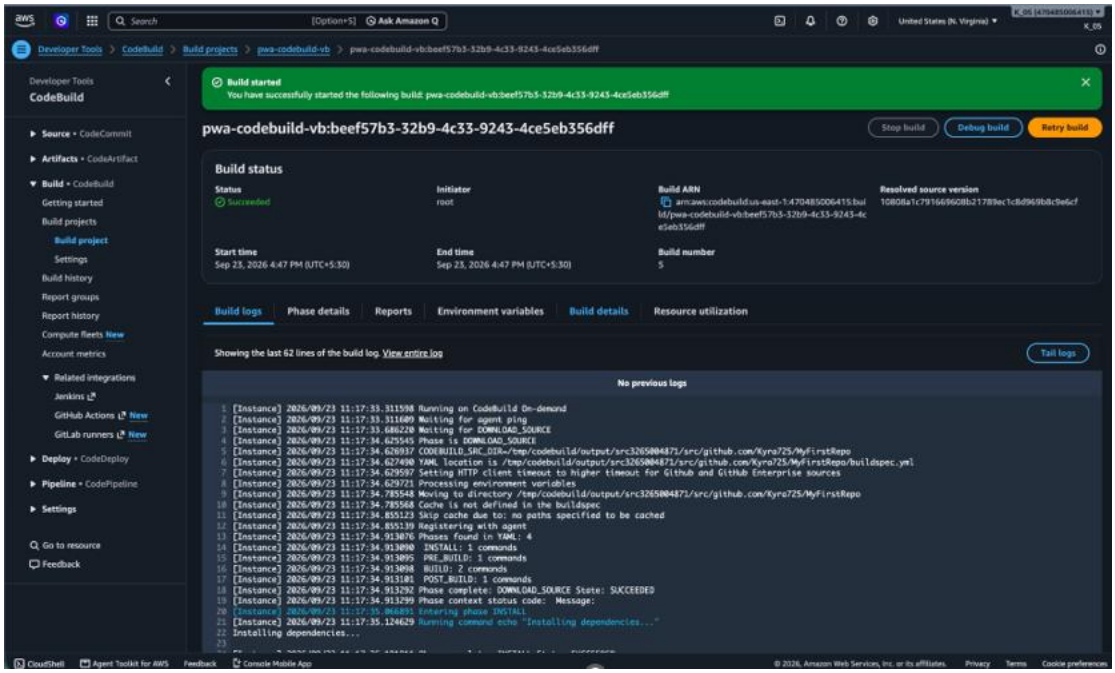
Build: AWS CodeBuild

Deployment: Amazon S3

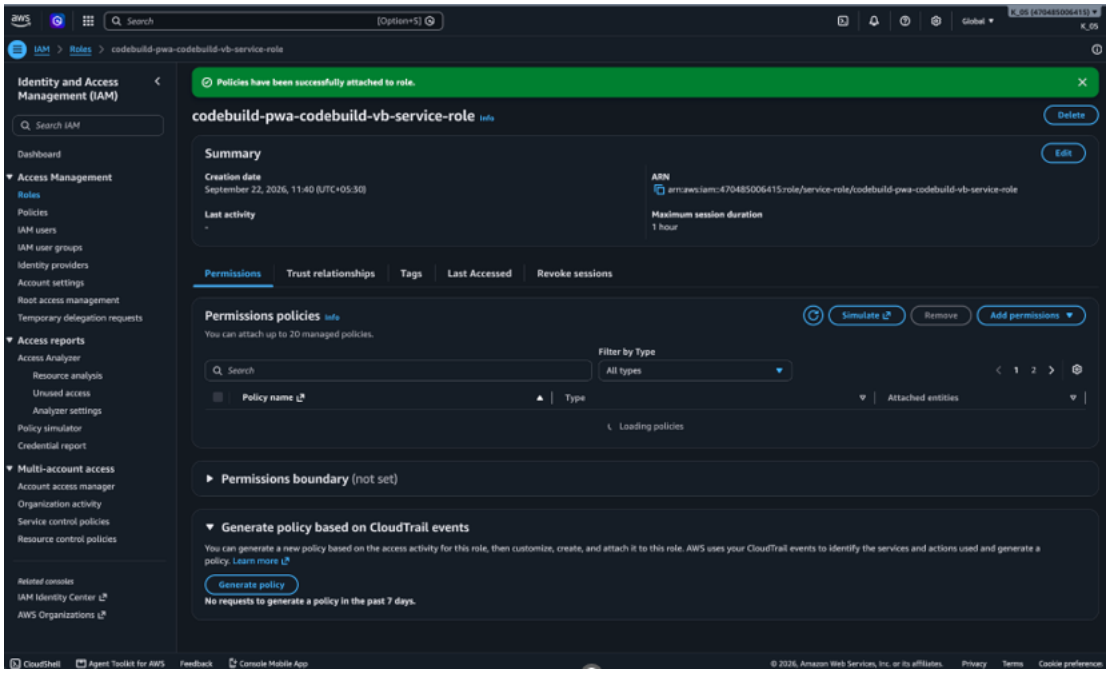


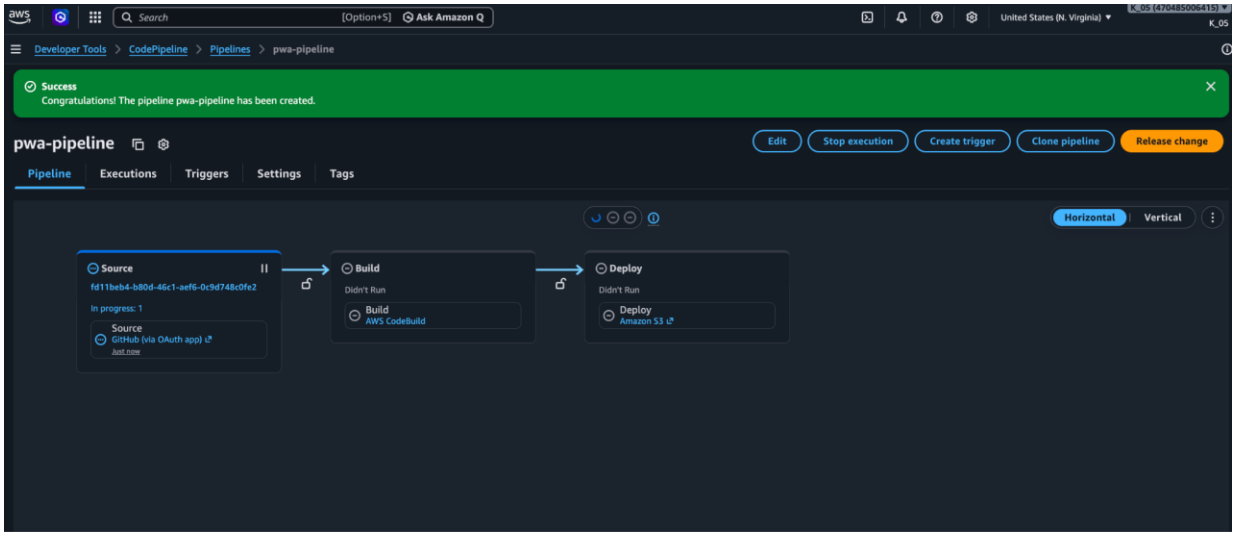
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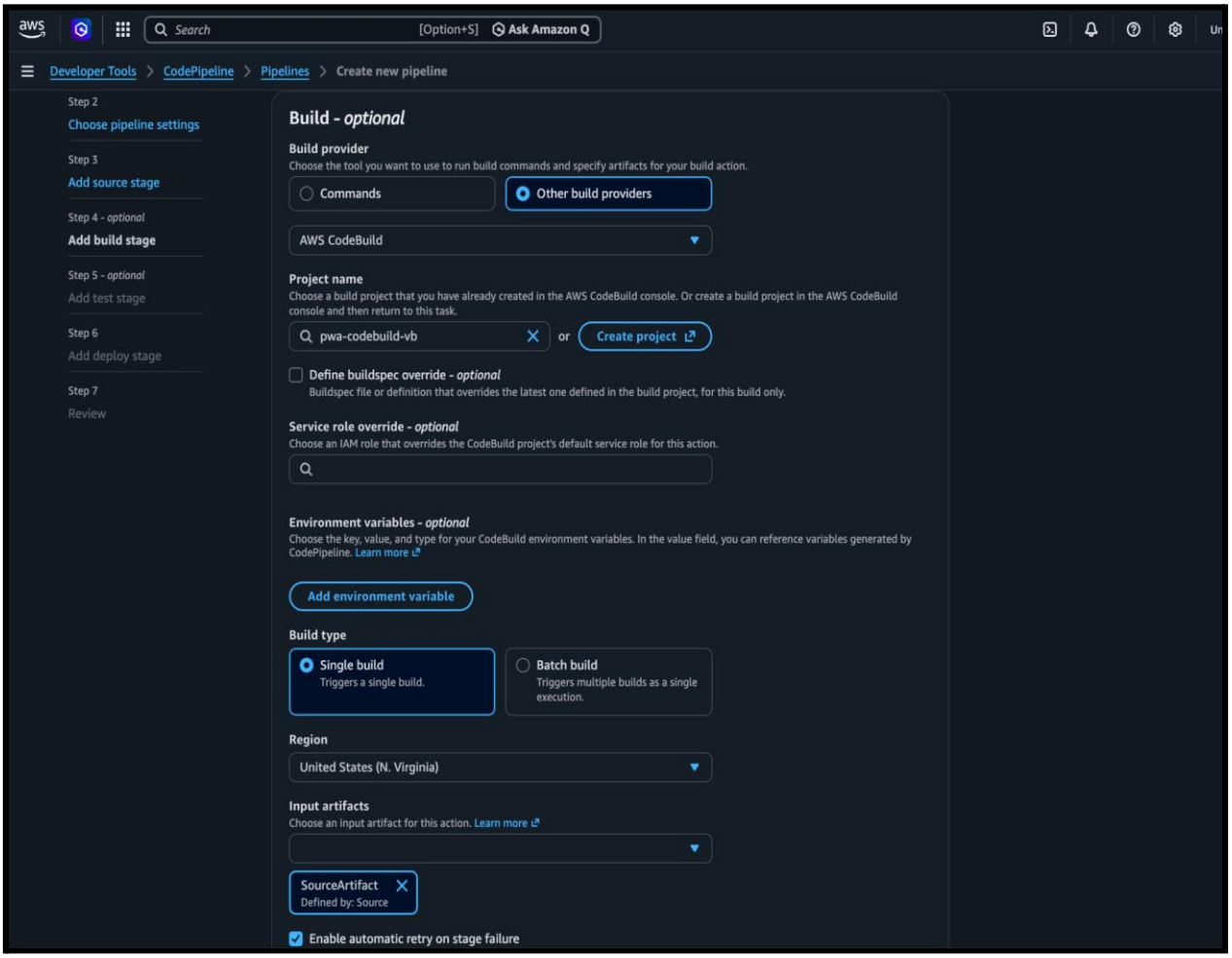


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Conclusion: Thus, I successfully implemented an AWS CI/CD pipeline using GitHub, AWS CodePipeline, AWS CodeBuild, and Amazon S3. I learned how to connect a GitHub repository with AWS, create and configure a CodeBuild project using buildspec.yml, and automate the build and deployment process through CodePipeline. I also learned how to host the deployed application as a static website on Amazon S3 and how changes made to the GitHub code can be automatically reflected on the deployed website.

LO4, LO6 mapped successfully.